Exp NO : 01

Setting Up the Environment And Preprocessing The Data

CODE :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3

d = pd.read_csv("/content/unemployment.csv")

d

d.head()

d.tail()

d.shape

d.columns

d.info()

d.describe()

conn = sqlite3.connect("/content/unemployment.db")

conn

d.dropna()

d.isnull().sum()
```

#OUTPUT 1:

	Country Name	Country Code	1991	1992	1993	1994	1995	1996	1997	1998	...	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
0	Africa Eastern and Southern	AFE	7.80	7.84	7.85	7.84	7.83	7.84	7.86	7.81	...	6.56	6.45	6.41	6.49	6.61	6.71	6.73	6.91	7.56	8.11
1	Afghanistan	AFG	10.65	10.82	10.72	10.73	11.18	10.96	10.78	10.80	...	11.34	11.19	11.14	11.13	11.16	11.18	11.15	11.22	11.71	13.28
2	Africa Western and Central	AFW	4.42	4.53	4.55	4.54	4.53	4.57	4.60	4.66	...	4.64	4.41	4.69	4.63	5.57	6.02	6.04	6.06	6.77	6.84
3	Angola	AGO	4.21	4.21	4.23	4.16	4.11	4.10	4.09	4.07	...	7.35	7.37	7.37	7.39	7.41	7.41	7.42	7.42	8.33	8.53
4	Albania	ALB	10.31	30.01	25.26	20.84	14.61	13.93	16.88	20.05	...	13.38	15.87	18.05	17.19	15.42	13.62	12.30	11.47	13.33	11.82
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
230	Samoa	WSM	2.10	2.38	2.63	3.04	3.19	3.47	3.90	4.18	...	8.75	8.67	8.72	8.50	8.31	8.58	8.69	8.41	9.15	9.84
231	Yemen, Rep.	YEM	8.32	8.31	8.35	8.34	8.96	9.59	10.20	10.81	...	13.17	13.27	13.47	13.77	13.43	13.30	13.15	13.06	13.39	13.57
232	South Africa	ZAF	29.95	29.98	29.92	29.89	29.89	29.87	29.91	29.95	...	24.73	24.56	24.89	25.15	26.54	27.04	26.91	28.47	29.22	33.56
233	Zambia	ZMB	18.90	19.37	19.70	18.43	16.81	15.30	13.64	12.00	...	7.85	8.61	9.36	10.13	10.87	11.63	12.01	12.52	12.85	13.03
234	Zimbabwe	ZWE	4.94	4.99	4.97	4.96	5.63	6.25	6.93	6.46	...	5.15	4.98	4.77	4.78	4.79	4.78	4.80	4.83	5.35	5.17

235 rows × 33 columns

#OUTPUT 2:

	Country Name	Country Code	1991	1992	1993	1994	1995	1996	1997	1998	...	2012	2013	2014	2015	2016	2017	2018	2019	2020	
0	Africa Eastern and Southern	AFE	7.80	7.84	7.85	7.84	7.83	7.84	7.86	7.81	...	6.56	6.45	6.41	6.49	6.61	6.71	6.73	6.91	7.56	
1	Afghanistan	AFG	10.65	10.82	10.72	10.73	11.18	10.96	10.78	10.80	...	11.34	11.19	11.14	11.13	11.16	11.18	11.15	11.22	11.71	1
2	Africa Western and Central	AFW	4.42	4.53	4.55	4.54	4.53	4.57	4.60	4.66	...	4.64	4.41	4.69	4.63	5.57	6.02	6.04	6.06	6.77	
3	Angola	AGO	4.21	4.21	4.23	4.16	4.11	4.10	4.09	4.07	...	7.35	7.37	7.37	7.39	7.41	7.41	7.42	7.42	8.33	
4	Albania	ALB	10.31	30.01	25.26	20.84	14.61	13.93	16.88	20.05	...	13.38	15.87	18.05	17.19	15.42	13.62	12.30	11.47	13.33	1

5 rows × 33 columns

#OUTPUT 3 & 4:

	Country Name	Country Code	1991	1992	1993	1994	1995	1996	1997	1998	...	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
230	Samoa	WSM	2.10	2.38	2.63	3.04	3.19	3.47	3.90	4.18	...	8.75	8.67	8.72	8.50	8.31	8.58	8.69	8.41	9.15	9.84
231	Yemen, Rep.	YEM	8.32	8.31	8.35	8.34	8.96	9.59	10.20	10.81	...	13.17	13.27	13.47	13.77	13.43	13.30	13.15	13.06	13.39	13.57
232	South Africa	ZAF	29.95	29.98	29.92	29.89	29.89	29.87	29.91	29.95	...	24.73	24.56	24.89	25.15	26.54	27.04	26.91	28.47	29.22	33.56
233	Zambia	ZMB	18.90	19.37	19.70	18.43	16.81	15.30	13.64	12.00	...	7.85	8.61	9.36	10.13	10.87	11.63	12.01	12.52	12.85	13.03
234	Zimbabwe	ZWE	4.94	4.99	4.97	4.96	5.63	6.25	6.93	6.46	...	5.15	4.98	4.77	4.78	4.79	4.78	4.80	4.83	5.35	5.17

5 rows × 33 columns

```
(235, 33)
```

#OUTPUT 5:

```
Index(['Country Name', 'Country Code', '1991', '1992', '1993', '1994', '1995',
       '1996', '1997', '1998', '1999', '2000', '2001', '2002', '2003', '2004',
       '2005', '2006', '2007', '2008', '2009', '2010', '2011', '2012', '2013',
       '2014', '2015', '2016', '2017', '2018', '2019', '2020', '2021'],
      dtype='object')
```

#OUTPUT 6:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 235 entries, 0 to 234
Data columns (total 33 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Country Name   235 non-null    object
 1   Country Code   235 non-null    object
 2   1991           235 non-null    float64
 3   1992           235 non-null    float64
 4   1993           235 non-null    float64
 5   1994           235 non-null    float64
 6   1995           235 non-null    float64
 7   1996           235 non-null    float64
 8   1997           235 non-null    float64
 9   1998           235 non-null    float64
 10  1999           235 non-null    float64
 11  2000           235 non-null    float64
 12  2001           235 non-null    float64
 13  2002           235 non-null    float64
 14  2003           235 non-null    float64
 15  2004           235 non-null    float64
 16  2005           235 non-null    float64
 17  2006           235 non-null    float64
 18  2007           235 non-null    float64
 19  2008           235 non-null    float64
 20  2009           235 non-null    float64
 21  2010           235 non-null    float64
 22  2011           235 non-null    float64
```

#OUTPUT 7:

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	...	2012	2013	2014	2015
count	235.000000	235.000000	235.000000	235.000000	235.000000	235.000000	235.000000	235.000000	235.000000	235.000000	...	235.000000	235.000000	235.000000	235.000000
mean	7.278000	7.626340	8.070766	8.246043	8.333915	8.494894	8.394043	8.441064	8.568043	8.438979	...	8.062553	8.086468	7.924340	7.818426
std	6.013749	6.296617	6.335855	6.243778	6.330822	6.358431	6.206845	6.133045	6.088361	6.126318	...	5.780173	5.832019	5.699899	5.574759
min	0.300000	0.340000	0.410000	0.470000	0.500000	0.560000	0.540000	0.560000	0.570000	0.580000	...	0.480000	0.250000	0.200000	0.170000
25%	2.945000	3.140000	3.700000	3.890000	3.945000	3.995000	4.020000	4.085000	4.275000	4.070000	...	4.090000	4.245000	4.200000	4.315000
50%	5.410000	5.710000	6.030000	6.550000	6.700000	7.050000	6.930000	6.890000	6.690000	6.530000	...	6.450000	6.290000	6.150000	6.080000
75%	9.815000	10.170000	10.895000	11.110000	11.050000	11.405000	11.090000	11.500000	11.845000	11.565000	...	10.655000	10.465000	10.290000	10.080000
max	36.120000	36.390000	36.740000	36.980000	37.340000	38.800000	37.940000	37.160000	36.350000	35.460000	...	31.020000	29.000000	28.030000	27.690000

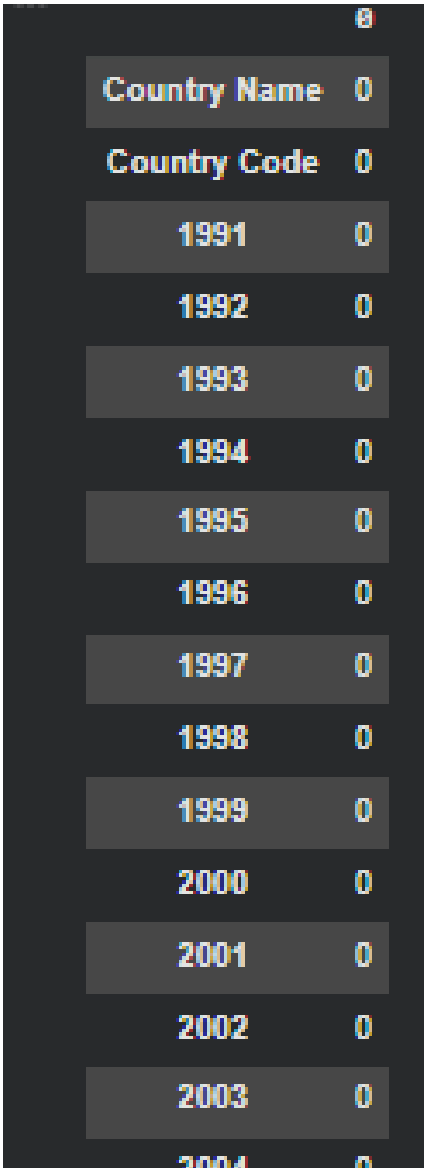
8 rows × 31 columns

#OUTPUT 8:

	Country Name	Country Code	1991	1992	1993	1994	1995	1996	1997	1998	...	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
0	Africa Eastern and Southern	AFE	7.80	7.84	7.85	7.84	7.83	7.84	7.86	7.81	...	6.56	6.45	6.41	6.49	6.61	6.71	6.73	6.91	7.56	8.11
1	Afghanistan	AFG	10.65	10.82	10.72	10.73	11.18	10.96	10.78	10.80	...	11.34	11.19	11.14	11.13	11.16	11.18	11.15	11.22	11.71	13.28
2	Africa Western and Central	AFW	4.42	4.53	4.55	4.54	4.53	4.57	4.60	4.66	...	4.64	4.41	4.69	4.63	5.57	6.02	6.04	6.06	6.77	6.84
3	Angola	AGO	4.21	4.21	4.23	4.16	4.11	4.10	4.09	4.07	...	7.35	7.37	7.37	7.39	7.41	7.41	7.42	7.42	8.33	8.53
4	Albania	ALB	10.31	30.01	25.26	20.84	14.61	13.93	16.88	20.05	...	13.38	15.87	18.05	17.19	15.42	13.62	12.30	11.47	13.33	11.82
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
230	Samoa	WSM	2.10	2.38	2.63	3.04	3.19	3.47	3.90	4.18	...	8.75	8.67	8.72	8.50	8.31	8.58	8.69	8.41	9.15	9.84
231	Yemen, Rep.	YEM	8.32	8.31	8.35	8.34	8.96	9.59	10.20	10.81	...	13.17	13.27	13.47	13.77	13.43	13.30	13.15	13.06	13.39	13.57
232	South Africa	ZAF	29.95	29.98	29.92	29.89	29.89	29.87	29.91	29.95	...	24.73	24.56	24.89	25.15	26.54	27.04	26.91	28.47	29.22	33.56
233	Zambia	ZMB	18.90	19.37	19.70	18.43	16.81	15.30	13.64	12.00	...	7.85	8.61	9.36	10.13	10.87	11.63	12.01	12.52	12.85	13.03
234	Zimbabwe	ZWE	4.94	4.99	4.97	4.96	5.63	6.25	6.93	6.46	...	5.15	4.98	4.77	4.78	4.79	4.78	4.80	4.83	5.35	5.17

235 rows × 33 columns

#OUTPUT 9:


0
Country Name 0
Country Code 0
1991 0
1992 0
1993 0
1994 0
1995 0
1996 0
1997 0
1998 0
1999 0
2000 0
2001 0
2002 0
2003 0